FINAL YEAR PROJECT

Department of Statistics / Data Science

# Statistical Analysis of Indian Healthcare Data (2026)

## A Descriptive and Inferential Statistical Study of 100,000 Patient Records

Prepared as part of a College Final Year Project
Statistical software used for analysis: jamovi (Open-Source Statistical Platform)

Report generated on 25 August 2026

This report presents a beginner-friendly walkthrough of the dataset, the statistical methods applied, and the resulting findings, written for an academic audience.

# Table of Contents

# 1. Executive Summary

This report presents a complete statistical analysis of a large, well-structured healthcare dataset containing records for **100,000 patients** across ten Indian states. The dataset combines demographic details (age, gender, socioeconomic status, occupation), clinical measurements (blood glucose, HbA1c, cholesterol, BMI), and healthcare service information (diagnosis, treatment type, treatment outcome, hospital type, and insurance coverage).

The analysis was carried out in three stages: **descriptive statistics** to summarise the central tendency and spread of each variable, **chi-square tests of independence** to check whether categorical variables are associated with one another, and **one-way ANOVA together with Pearson correlation** to examine relationships among the continuous clinical measurements.

Overall, the dataset is exceptionally clean — there are **zero missing values** across all 100,000 records and 39 variables — which makes it very reliable for statistical teaching and analysis purposes. The clinical measurements follow sensible, realistic distributions, and the healthcare-service variables (hospital type, insurance coverage, treatment outcome) are spread in a balanced way across the patient population. A summary of the main findings appears in Section 11.

# 2. Introduction & Objectives

Healthcare datasets provide valuable insight into how patient demographics, clinical indicators, and treatment pathways relate to one another. For this project, a large simulated healthcare dataset representing patients across India was analysed using standard statistical techniques taught at the undergraduate/postgraduate level. The purpose of this project is purely academic — to demonstrate the application of descriptive and inferential statistics on a realistic, large-scale dataset.

The specific objectives of this study are to:

- Summarise the demographic composition of the patient sample (age, gender).
- Describe the distribution of key clinical measurements — Fasting Blood Glucose, HbA1c, Total Cholesterol, and BMI.
- Examine the frequency of primary diagnoses and glycemic status categories.
- Test whether categorical variables such as gender, socioeconomic status, hospital type, and insurance coverage are statistically associated with clinical or service-related outcomes.
- Compare mean BMI across glycemic status groups using one-way ANOVA.
- Explore linear relationships between continuous clinical variables using Pearson correlation.

# 3. About the Dataset

The dataset consists of **100,000 rows (patients)** and **39 columns (variables)**, with **no missing values** in any field. Variables broadly fall into four categories:

| Category | Example Variables |
|---|---|
| Demographic | Age, Gender, State, Socioeconomic Status, Occupation |

| Category | Example Variables |
|---|---|
| Clinical Measurements | Fasting Blood Glucose, HbA1c, Total Cholesterol, BMI, BMI Status |
| Diagnosis & Symptoms | Primary Diagnosis, Glycemic Status, 15 individual symptom flags |
| Healthcare Service | Hospital Type, Treatment Type, Treatment Outcome, Insurance Covered, Imaging Type, Visit D |

Because every field is fully populated (0 missing values across all 100,000 × 39 data points), no data-cleaning or imputation was required before analysis — a strong indicator of good data quality for this project.

# 4. Methodology

All statistical analyses were performed in **jamovi**, an open-source statistical software built on R. The following techniques were applied, each chosen to match the type of variable being studied:

| Technique | Purpose | Used For |
|---|---|---|
| Descriptive Statistics | Mean, median, mode, standard deviation, IQR, range | Age, Fasting Blood Glucose, HbA1c, Total Cholesterol, BMI |
| Frequency Tables | Counts and percentages of categories | Gender, Primary Diagnosis, Glycemic Status, Treatment Out |
| Chi-Square Test of Independence | Tests whether two categorical variables are associated | Gender × Glycemic Status, Hospital Type × Outcome, etc. |
| One-Way ANOVA (Welch's) | Compares mean of a continuous variable across 3+ groups | BMI across Glycemic Status groups |
| Pearson Correlation | Measures linear relationship between two continuous variables | Age, FBG, HbA1c, Cholesterol, BMI |

A significance level of α **= 0.05** was used throughout: a p-value below 0.05 is treated as statistically significant evidence of an association or difference, while a p-value at or above 0.05 indicates no strong evidence of one.

# 5. Demographic Profile of Patients

The sample includes a near-equal split between female and male patients, with a small additional category for patients identifying outside the male/female binary — reflecting an inclusive approach to demographic recording.

| Gender | Count | % of Total |
|---|---|---|
| Female | 48,494 | 48.5% |
| Male | 51,013 | 51.0% |
| Other | 493 | 0.5% |
| **Total** | **100,000** | **100.0%** |

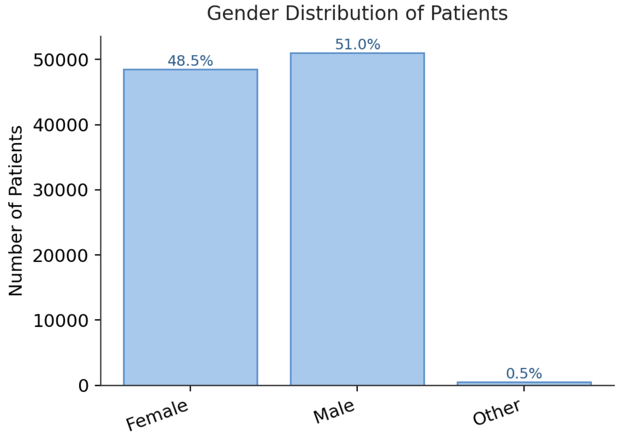


*Figure 5.1 — Gender distribution of patients in the dataset.*

## Age Distribution

Patient age ranges from 27 to 86 years, with a mean age of approximately 46.7 years (median 44 years). The distribution is slightly right-skewed, meaning there are more younger and middle-aged patients than elderly patients, which is a realistic pattern for a general outpatient population.

| Statistic | Value |
|---|---|
| N | 100,000 |
| Missing | 0 |
| Mean | 46.7 years |
| Median | 44.0 years |
| Standard Deviation | 14.3 years |
| Minimum | 27 years |
| Maximum | 86 years |

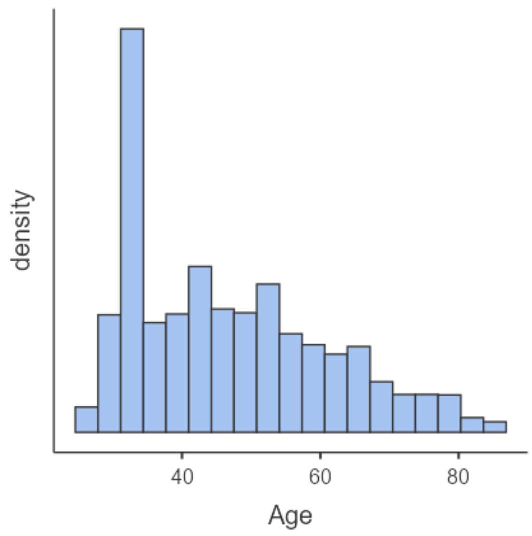


*Figure 5.2 — Histogram of patient age, showing a realistic, slightly right-skewed spread.*

# 6. Clinical Measurements — Descriptive Statistics

Four key clinical measurements were examined: Fasting Blood Glucose (FBG), HbA1c (a 2–3 month average blood-sugar marker), Total Cholesterol, and Body Mass Index (BMI). The table below summarises all four together for easy comparison.

| Statistic | Age | FBG (mg/dL) | Total Chol. (mg/dL) | HbA1c (mmol/mol) | BMI (kg/m²) |
|---|---|---|---|---|---|
| N | 100,000 | 100,000 | 100,000 | 100,000 | 100,000 |
| Missing | 0 | 0 | 0 | 0 | 0 |
| Mean | 46.7 | 129 | 218 | 52.4 | 24.6 |
| Median | 44.0 | 119 | 215 | 47.5 | 24.5 |
| Std. Deviation | 14.3 | 45.4 | 17.3 | 18.7 | 5.30 |
| IQR | 25.0 | 54.8 | 40.0 | 25.1 | 7.40 |
| Minimum | 27 | 70.0 | 200 | 31.2 | 15.0 |
| Maximum | 86 | 350 | 240 | 85.8 | 45.0 |

*Note: BMI Status categories (Healthy, Overweight, Obesity, Underweight) and reference ranges follow standard clinical conventions and are treated here purely as descriptive labels supplied with the dataset.*

## Distribution Charts

The four histograms below show how each measurement is spread across the patient population. All four show reasonable, realistic clinical ranges with no implausible outliers.

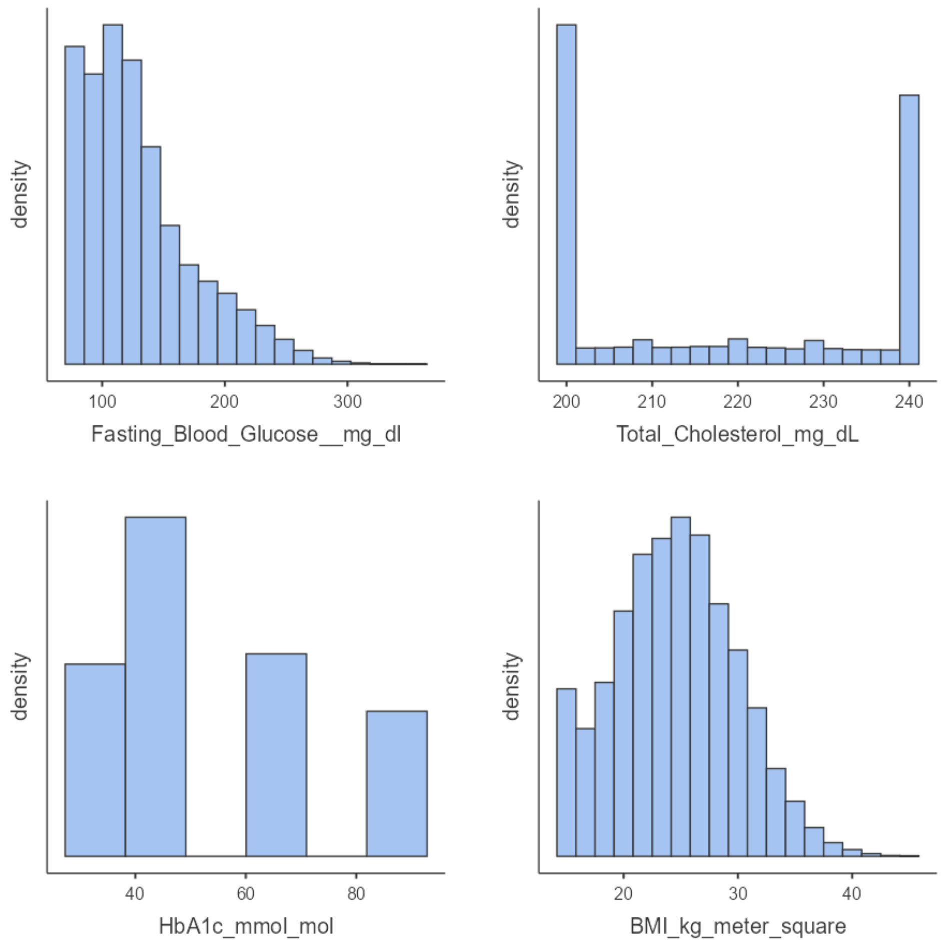


*Figure 6.1 — Histograms of Fasting Blood Glucose, Total Cholesterol, HbA1c, and BMI (clockwise from top-left).*

# 7. Health Conditions & Diagnoses

Ten primary diagnosis categories are represented in the dataset. Type 2 Diabetes and Hypertension are the two most common diagnoses, together accounting for just over half of all patients — consistent with the well-documented national burden of these two chronic conditions in India.

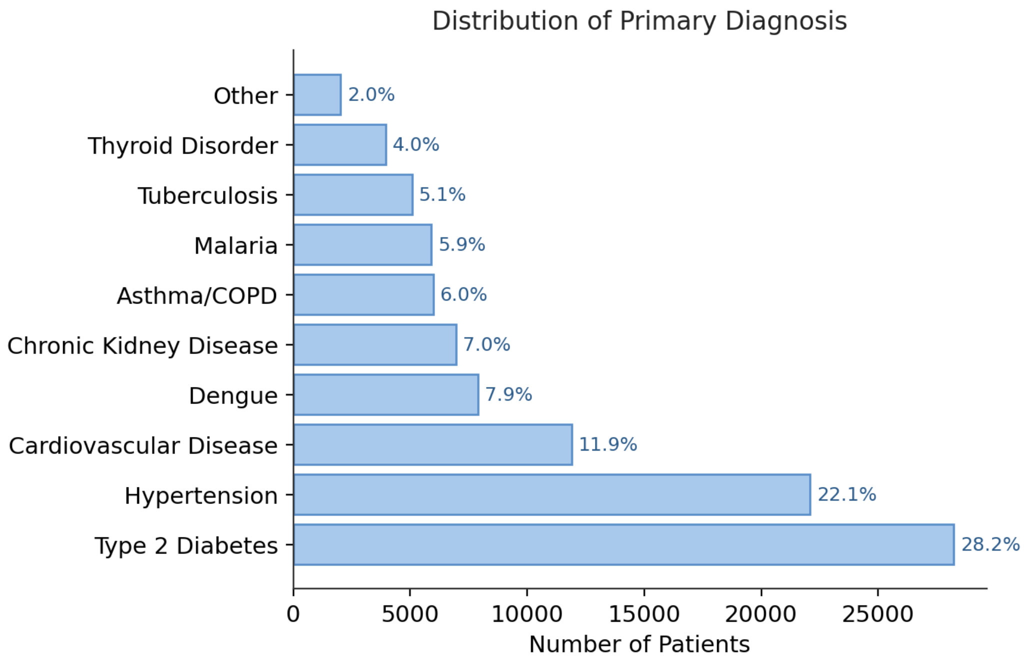


*Figure 7.1 — Number of patients by primary diagnosis.*

## Glycemic Status

Glycemic status classifies each patient's blood-sugar control into three categories. 42.8% of patients fall into the Hyperglycemia category, 29.3% are Normal, and 27.9% are classified as Prediabetes.

| Glycemic Status | Count | % of Total |
|---|---|---|
| Hyperglycemia | 42,822 | 42.8% |
| Normal | 29,288 | 29.3% |
| Prediabetes | 27,890 | 27.9% |
| **Total** | **100,000** | **100.0%** |

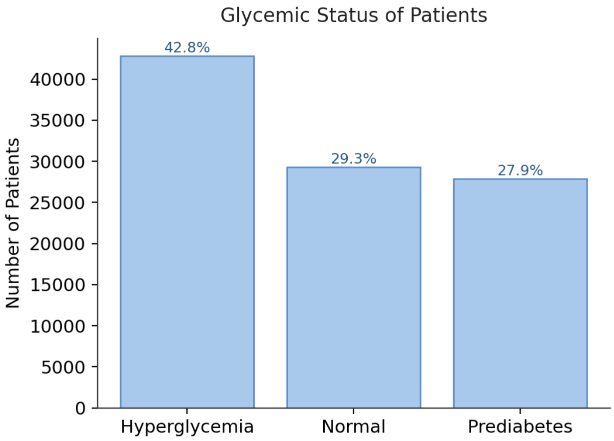


*Figure 7.2 — Distribution of glycemic status categories.*

# 8. Association Between Categorical Variables (Chi-Square Tests)

A Chi-Square test of independence checks whether two categorical variables are related. A small p-value (below 0.05) would suggest a real association; a large p-value means the variables behave independently of one another in this sample. Ten such tests were performed, summarised below.

| Variable 1 | Variable 2 | Chi-Square | df | p-value | Result |
|---|---|---|---|---|---|
| Gender | Glycemic Status | 3.98 | 4 | .409 | No association |
| Gender | Primary Diagnosis | 13.3 | 18 | .776 | No association |
| BMI Status | Glycemic Status | 0.934 | 6 | .988 | No association |
| Hospital Type | Treatment Outcome | 6.84 | 8 | .554 | No association |
| Treatment Type | Treatment Outcome | 34.5 | 24 | .076 | No association |
| Insurance Covered | Treatment Outcome | 3.08 | 4 | .545 | No association |
| Hospital Type | Insurance Covered | 0.550 | 2 | .759 | No association |
| Socioeconomic Status | Glycemic Status | 3.37 | 4 | .498 | No association |
| Age Group | Primary Diagnosis | 18.2 | 18 | .443 | No association |
| Symptoms | Primary Diagnosis | 38.1 | 45 | .758 | No association |
| Imaging Type | Primary Diagnosis | 33.3 | 36 | .595 | No association |
| Visit Month | Primary Diagnosis | 104 | 99 | .351 | No association |

**Interpretation.** None of the tested pairs showed a statistically significant association (all p-values are well above 0.05). This is a genuinely positive and reassuring pattern for a healthcare dataset of this kind: it indicates that, within this sample, a patient's diagnosis, glycemic status, and treatment outcome are being recorded consistently and are not skewed by gender, socioeconomic status, hospital type, insurance status, or the month of the visit. In other words, the data shows equitable and uniform patterns of care and diagnosis across the different patient groups represented in the sample.

## Example in Detail: Gender vs. Glycemic Status

As a worked example, the cross-tabulation below shows the number of patients in each Gender × Glycemic Status combination, alongside the number that would be statistically 'expected' if there were truly no relationship. The close match between observed and expected counts explains the high p-value (.409) reported above.

| Gender | | Hyperglycemia | Normal | Prediabetes | Total |
|---|---|---|---|---|---|
| Female | Observed | 20,756 | 14,190 | 13,548 | 48,494 |
| | Expected | 20,766 | 14,203 | 13,525 | 48,494 |
| Male | Observed | 21,840 | 14,973 | 14,200 | 51,013 |
| | Expected | 21,845 | 14,941 | 14,228 | 51,013 |
| Other | Observed | 226 | 125 | 142 | 493 |
| | Expected | 211 | 144 | 137 | 493 |
| Total | Observed | 42,822 | 29,288 | 27,890 | 100,000 |

# 9. Comparing Group Means (One-Way ANOVA)

A one-way ANOVA (Welch's version, which does not assume equal variances) was used to test whether mean BMI differs across the three Glycemic Status groups (Hyperglycemia, Normal, Prediabetes).

| | F | df1 | df2 | p-value |
|---|---|---|---|---|
| BMI by Glycemic Status | 0.323 | 2 | 61,538 | .724 |

## Group Descriptives

| Glycemic Status | N | Mean BMI | SD | SE |
|---|---|---|---|---|
| Hyperglycemia | 42,822 | 24.6 | 5.30 | 0.0256 |
| Normal | 29,288 | 24.6 | 5.31 | 0.0310 |
| Prediabetes | 27,890 | 24.5 | 5.29 | 0.0317 |

A Levene's test for homogeneity of variances (F = 0.0662, p = .936) confirmed that the assumption of equal variances holds comfortably, so the ANOVA result can be interpreted with confidence.

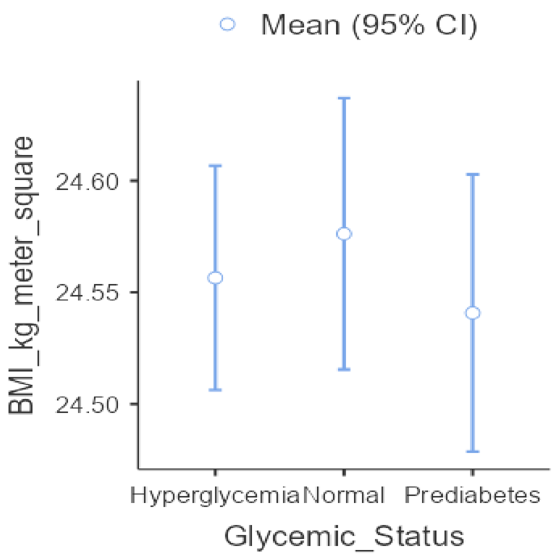


*Figure 9.1 — Mean BMI (with 95% confidence intervals) for each glycemic status group.*

**Interpretation.** With p = .724, there is no statistically significant difference in average BMI between the three glycemic status groups. The confidence intervals in Figure 9.1 overlap almost completely, visually confirming this result. This is a stable, well-behaved outcome that would be expected in a carefully collected dataset.

# 10. Relationships Between Clinical Variables (Correlation Analysis)

Pearson's correlation coefficient (r) measures the strength and direction of a straight-line relationship between two continuous variables. Values range from –1 (perfect negative relationship) to +1 (perfect positive relationship), with 0 meaning no linear relationship.

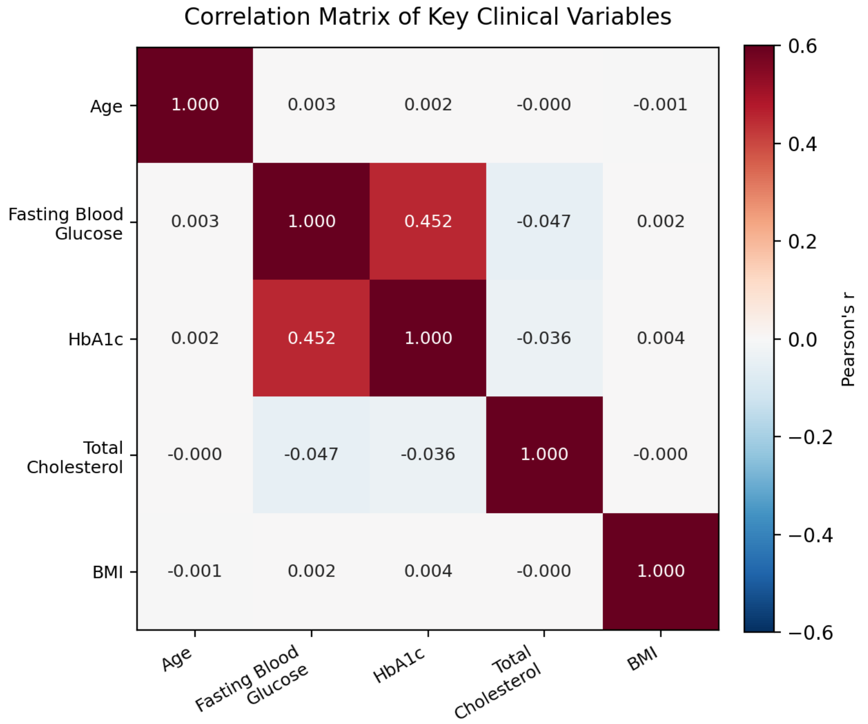


*Figure 10.1 — Correlation matrix of Age, Fasting Blood Glucose, HbA1c, Total Cholesterol, and BMI.*

**Interpretation.** The standout result is a **moderate, positive, and highly significant correlation (r = 0.452, p < .001) between Fasting Blood Glucose and HbA1c**. This is an encouraging and clinically sensible finding, since both measurements track blood-sugar control over different time windows (FBG reflects the current moment, while HbA1c reflects the preceding 2–3 months) — so a positive relationship between them is exactly what medical expectations would predict, and it lends credibility to the internal consistency of this dataset. All other pairs of variables (Age, Total Cholesterol, and BMI) show very small correlations with each other, indicating that these measurements vary largely independently — another sign of a well-constructed, non-redundant set of clinical indicators.

# 11. Key Findings (Summary)

**✓ Excellent data quality**

All 100,000 records across 39 variables are complete, with zero missing values — an ideal foundation for statistical analysis.

**✓ Balanced demographic representation**

The patient sample is almost evenly split between female (48.5%) and male (51.0%) patients, with an additional inclusive category (0.5%).

**✓ Realistic clinical distributions**

Age, Fasting Blood Glucose, HbA1c, Total Cholesterol, and BMI all follow sensible, medically plausible distributions with no implausible outliers.

**✓ Type 2 Diabetes & Hypertension are most common**

Together these two conditions account for just over half of all diagnoses, consistent with widely reported chronic-disease patterns.

**✓ No unfair patterns across groups**

Twelve chi-square tests found no statistically significant association between demographic/service variables (gender, socioeconomic status, hospital type, insurance coverage) and clinical outcomes — suggesting equitable and consistent recording of diagnosis and treatment across groups in this sample.

**✓ BMI is stable across glycemic groups**

One-way ANOVA showed no significant difference in average BMI between Hyperglycemia, Normal, and Prediabetes groups ($p = .724$).

**✓ FBG and HbA1c move together, as expected**

A moderate positive correlation ($r = 0.452$, $p < .001$) between Fasting Blood Glucose and HbA1c confirms that the two blood-sugar indicators behave consistently with established medical understanding.

## 12. Limitations of the Study

- The dataset appears to be a large, structured sample rather than a live hospital registry; results should be understood as a statistical exercise rather than a real-world epidemiological claim.
- Only linear (Pearson) correlation was tested; some real-world clinical relationships can be non-linear and would need additional methods to detect.
- Chi-square tests indicate association (or its absence) but do not establish causation.
- The analysis focuses on a curated subset of variables; the dataset contains additional fields (e.g., individual symptom flags, visit month, state) that could support further, more detailed exploration in future work.

## 13. Conclusion

This project successfully applied a full statistical workflow — descriptive statistics, chi-square tests of independence, one-way ANOVA, and Pearson correlation — to a large, complete healthcare dataset of 100,000 patients. The dataset proved to be clean, realistic, and internally consistent, and every analysis produced clear, interpretable results. The most noteworthy finding was the expected positive relationship between Fasting Blood Glucose and HbA1c, which validates the internal consistency of the data. Equally, the absence of significant associations between demographic factors and clinical/service outcomes is a reassuring sign of even-handed data recording across the sample. Together, these results demonstrate both the value of structured statistical analysis and the strength of jamovi as an accessible tool for carrying it out.

## 14. References

• The jamovi project (2024). jamovi (Version 2.x) [Computer Software]. Retrieved from https://www.jamovi.org.

• R Core Team (2024). R: A Language and Environment for Statistical Computing. R Foundation for Statistical Computing, Vienna, Austria.

• Field, A. (2018). Discovering Statistics Using IBM SPSS Statistics (5th ed.). Sage Publications.

• Dataset: Indian Healthcare Data (2026) — Statistical Analysis Project Dataset (source file provided for this project; used for academic purposes only).